



LAB EXPERIMENTS

ECA1011 Microprocessors and Microcontrollers

Submitted To:

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DATE: 15/10/2026

EXPERIMENT 01: ANALYSIS, DEVELOPMENT AND VERIFICATION OF 8-BIT AND 16-BIT ARITHMETIC OPERATIONS USING ASSEMBLY LANGUAGE PROGRAM

Objective:

1. To analyze, develop & verify 8-bit Addition and Subtraction
2. To analyze, develop & verify 16-bit Addition and Subtraction
3. To analyze, develop & verify 8-bit Multiplication and Division
4. To analyze, develop & verify 16-bit Multiplication and Division

Components Required:

8085 Simulator or 8085 Intel Microprocessor

Procedure:

i. Switch ON the 8085-microprocessor kit or open the 8085 simulator.
ii. Enter the program opcodes into the specified starting memory address using the memory examine/substitute command.
iii. Store the required input data in the specified memory locations, if provided.
iv. Execute the program from the given starting address using the GO/EXEC command.
v. Examine the memory locations assigned for the results.
vi. Record the output values and verify them with the expected arithmetic results.
vii. Repeat the procedure for all four programs and record the observations.

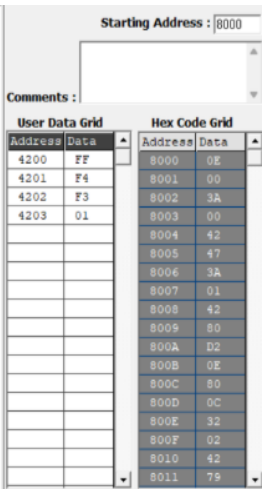
1. Analysis of 8-bit Addition and Subtraction :

PROGRAM [Additiom]

```
MVI C 00h
LDA 4200h
MOV B A
LDA 4201h
ADD B
JNC SKIP
INR C
SKIP : STA 4202h
MOV A C
STA 4203h
HLT
```

OUTPUT

```
PRIYANKA G [192524124]
MVI C 00h
LDA 4200h
MOV B A
LDA 4201h
ADD B
JNC SKIP
INR C
SKIP: STA 4202h
MOV A C
STA 4203h
HLT
```



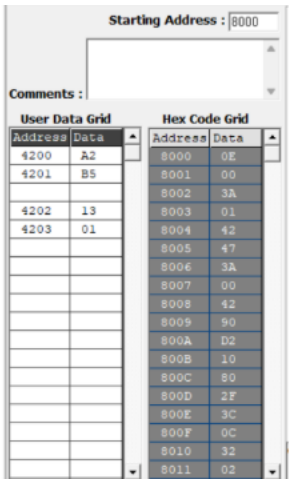
Memory	Data
4200	FF
4201	F4
4202	F3
4203	01

PROGRAM [Subtraction]

MVI C 00h
LDA 4201h
MOV B A
LDA 4200h
SUB B
JNC SKIP
CMA
INR A
INR C
SKIP : STA 4202h
MOV A C
STA 4203h
HLT

OUTPUT

```
PRIYANKA G [192524124]
MVI C 00h
LDA 4201h
MOV B A
LDA 4200h
SUB B
JNC SKIP
CMA
INR A
INR C
SKIP: STA 4202h
MOV A C
STA 4203h
HLT
```



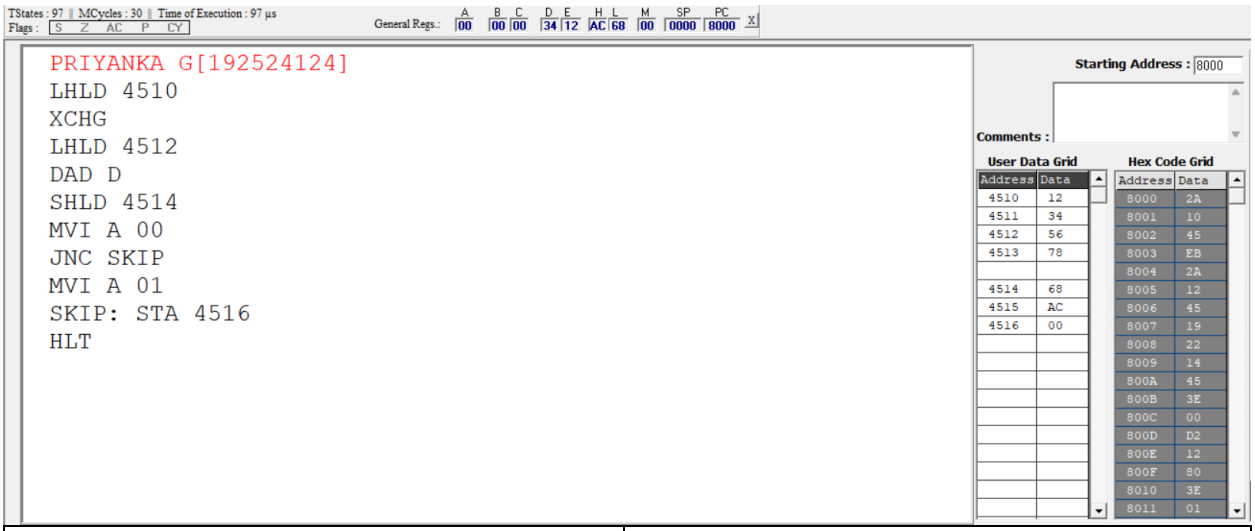
Memory	Data
4200	A2
4201	B5
4202	13
4203	01

2. Analysis of 16-bit Addition and Subtraction:

PROGRAM [Addition]

```
LHLD 4510
XCHG
LHLD 4512
DAD D
SHLD 4514
MVI A 00
JNC SKIP
MVI A 01
SKIP: STA 4516
HLT
```

OUTPUT

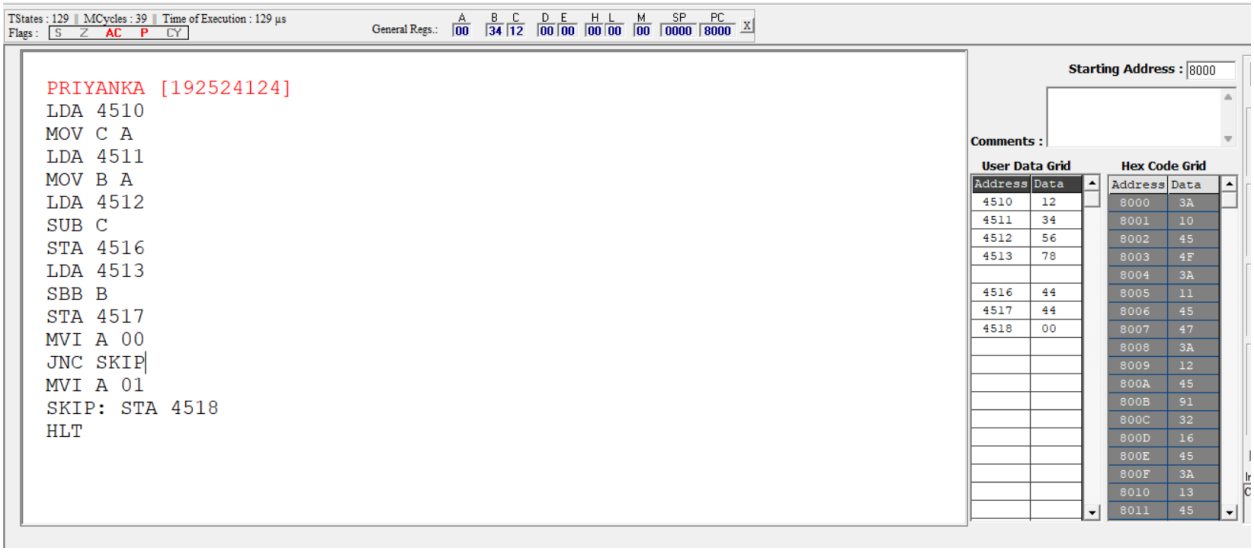


Memory	Data
4210	12
4211	34
4212	56
4213	78
4216	68
4217	AC
4218 (carry)	00

PROGRAM [Subtraction]

```
LDA 4510
MOV C  A
LDA 4511
MOV B  A
LDA 4512
SUB C
STA 4516
LDA 4513
SBB B
STA 4517
MVI A 00H
JNC SKIP
MVI A  01
SKIP: STA 4518
HLT
```

OUTPUT



Memory	Data
4210	12
4211	34
4212	56
4213	78
4216	44
4217	44
4218 (carry)	00

3. Analysis of 8-bit Multiplication and Division:

PROGRAM [Multiplication]

```
LXI H 0000H
LDA 4201h
MOV E A
LDA 4202
MOV C A
REPEAT : DAD D
DCR C
JNZ REPEAT
SHLD 4203
HLT
```

OUTPUT

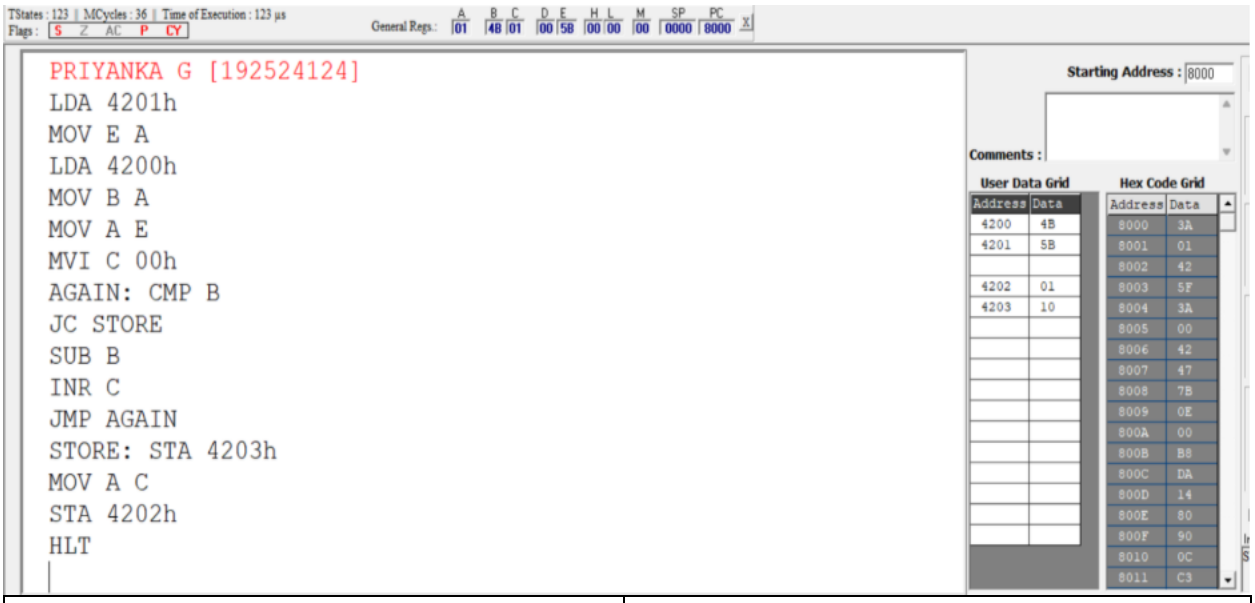


Memory	Data
4201	1A
4202	C4
4203	E8
4204	13

PROGRAM [Division]

LDA 4201h
MOV E A
LDA 4200h
MOV B A
MOV A E
MVI C 00h
AGAIN: CMP B
JC STORE
SUB B
INR C
JMP AGAIN
STORE: STA 4203h
MOV A C
STA 4202h
HLT

OUTPUT



Memory	Data
4200	4A
4201	5B
4202	01
4203	10

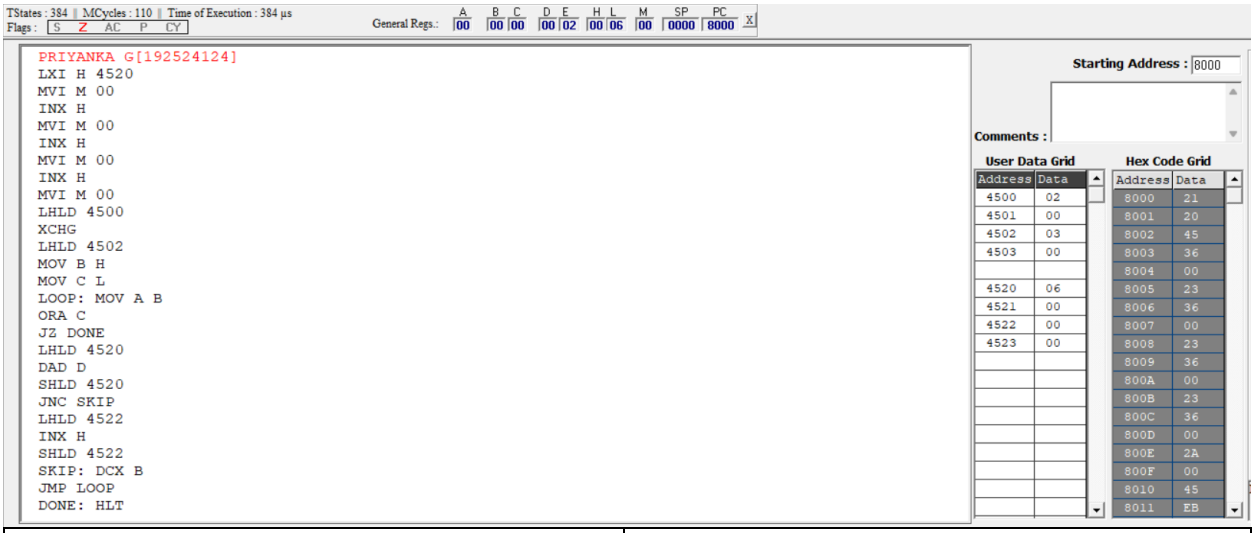
4. Analysis of 16-bit Multiplication and Division:

PROGRAM [Multiplication]

```
LXI H 4520
MVI M 00
INX H
MVI M 00
INX H
MVI M 00
INX H
MVI M 00
LHLD 4500
XCHG
LHLD 4502
MOV B H
MOV C L
LOOP: MOV A B
ORA C
JZ DONE
```

LHLD 4520
DAD D
SHLD 4520
JNC SKIP
LHLD 4522
INX H
SHLD 4522
SKIP: DCX B
JMP LOOP
DONE: HLT

OUTPUT



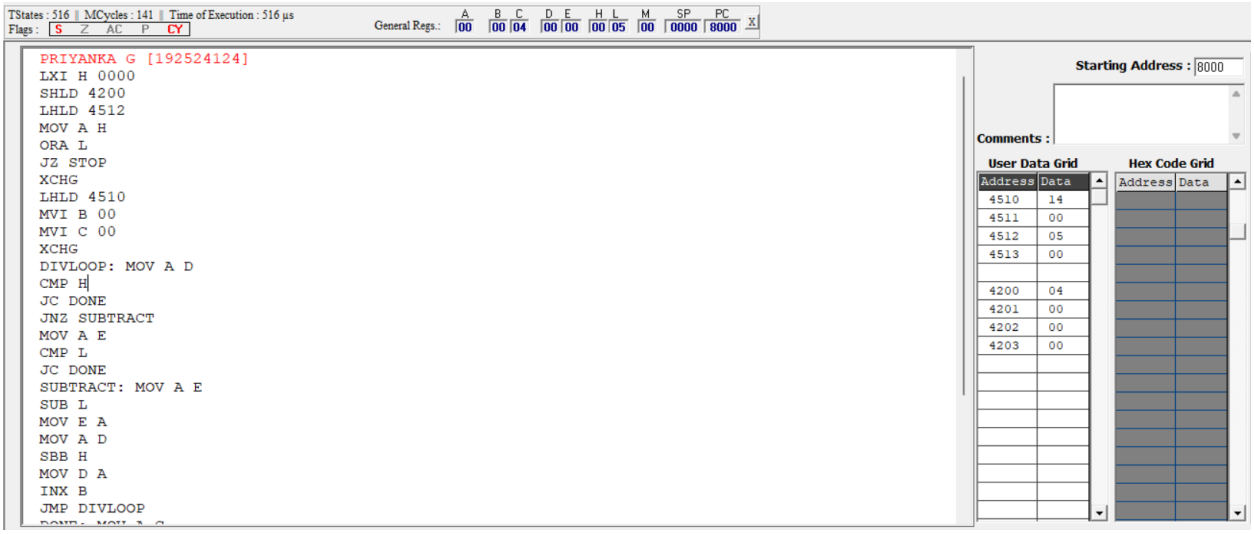
Memory	Data
4500	02
4501	00
4502	03
4503	00
4520	06
4521	00
4522	00
4523	00

PROGRAM [Division]

LXI H 0000
SHLD 4514

```
LHLD 4510
XCHG
DIVLOOP:
LHLD 4512
MOV A E
SUB L
MOV E A
MOV A D
SBB H
MOV D A
JC DONE
XCHG
SHLD 4510
XCHG
LDA 4514
INR A
STA 4514
JNZ DIVLOOP
LDA 4515
INR A
STA 4515
JMP DIVLOOP
DONE:
XCHG
SHLD 4516
HLT
```

OUTPUT



Memory	Data
4510	14
4511	00
4512	05
4513	00
4200	04
4201	00
4202	00
4203	00

Result:

Thus, 8-bit and 16-bit addition, subtraction, multiplication, and division programs were executed successfully using the 8085 microprocessors, and the results were verified by examining the specified memory locations and flag conditions.